

BIG DATA SYSTEMS – ASSIGNMENT 2

Amazon Product Review Analysis using Apache Spark

1. Abstract:

This assignment is designed to provide students with practical, hands-on experience in leveraging Apache Spark for large-scale data processing and analysis. By working with the Amazon Product Reviews dataset, a publicly available real-world corpus, students will gain exposure to Spark's diverse capabilities, including data ingestion, transformation, SQL-based querying, and advanced analytics. The assignment emphasizes not only technical proficiency but also the ability to derive meaningful insights from complex datasets. Through this exercise, students will strengthen their understanding of distributed computing concepts while applying Spark to solve realistic data challenges.

2. Dataset:

Amazon Product Reviews dataset (AmazonProductReviews.csv) for this assignment is attached in the Assignment section of Taxila.

The columns required for the analytical queries in this assignment are specified below:

CSV Column	Description
categories	Comma separated list of product categories.
name	Product name
reviews.date	Date of the review
reviews.rating	Rating given by reviewer (1-5)
reviews.text	Full text of the review
reviews.title	Title/headline of the review
reviews.username	Username of the reviewer

3. Assignment Questions

- i. Data loading: Load the dataset into a Spark with schema inference as appropriate. Print the schema and the number of records loaded.
- ii. Data cleansing: Make necessary changes in Schema, as required for the given analytical queries. Create a new column with name "primary_category" and populate it with the first category present in the 'categories' column. Drop rows where rating is missing or outside the valid range (1–5). List the modified schema and find the number of records after dropping the rows.
- iii. Top Products by Average Rating with Minimum Reviews: Find product names with at least 20 reviews and rank them by average rating.
- iv. Most Active Reviewers (by Review Count): List top 10 most active reviewers.
- v. Show how ratings evolve over time per category: Show the Monthly Trend of Average Ratings per primary_category of product. Show 40 rows.
- vi. Find products loved by some and hated by few: List top 10 Product names by the Ratio of 5-Star to 1-Star Reviews.
- vii. Longest Review Texts per Category: List the longest review in each primary category along with review title and length of review text. List it in the order of review length. For this question, result can be displayed with "truncate=True" as the reviews can be very long.
- viii. Growth in review volume: Show the Year-over-Year Growth in Review Counts.
- ix. Average Rating by Review Length buckets: Analyzes whether longer reviews tend to be more positive or negative. Use 3 buckets as given below:

Short: length of review text is less than 50.

Medium: length of review text is between 50 and 200 (both inclusive)

Long: length of review text is above 200.

- x. Products with Declining Ratings Over Time: Identify the top 10 product names whose ratings have dropped maximum. Use monthly average rating (first and last averages) of a product to identify its rating drop.
- xi. Products with Declining Ratings Over Time - Analysis: From on the results of the above question (top products whose ratings dropped maximum), select one product and do a data-driven analysis of the decline and provide your finding and recommendations (max 150 words).
- xii. After implementing all the above analytical queries, identify a query where there is a performance bottleneck in your Spark job and propose and implement optimizations. Compare the execution times before and after optimization, and justify your approach.

4. Submission Requirements:

Deliverables:

1. Report (PDF/Word):

- Document the following:
 - a. Group members and contribution percentage
 - b. Problem-solving approach.
 - c. Details of the development environment and setup.
 - d. Code/query and results of each query/question.
 - e. Justifications for key decisions

Note: *The primary evaluation will be based on the submitted report. Therefore, it must be comprehensive and include all required details. For each question, relevant code along with its corresponding output should be clearly documented. Outputs should be easily readable—either presented in well-formatted tables, clear images of execution results, or preferably a combination of both.*

2. Jupyter Notebook:

- For this assignment, you will leverage Apache Spark with PySpark and SparkSQL to analyze the dataset. You can choose to work with either Google Colab or a local Jupyter Notebook. If using the latter, ensure Spark is installed and properly configured on the local machine.
- Ensure high code quality with inline documentation.

- Submit the notebook file (.ipynb) containing outputs for all cells.

3. Video (mp4):

- Record the data loading and wrangling steps and execution of queries and their outcomes in real-time as a .mp4 file.

Submission File Format:

- Submit the three files specified in the 'Deliverables' section viz. the Jupyter notebook(.ipynb) file, the report (.docx/.pdf) file and the video file
 - Report: **Asgn2_Grp_<your_group_no>_report.pdf** OR **Asgn2_Grp_<your_group_no>_report.docx**
 - Notebook: **Asgn2_Grp_<your_group_no>_code.ipynb**
 - Video: **Asgn2_Grp_<your_group_no>_video.mp4**

General Notes:

- ☐ The report document should have full name and the BITS Registration number of each group member.
- ☐ Each group consists of 3 students. For the assignment, one of the members may act as the group leader.
 - The Group leader will be providing a contribution weightage of all group members and marks will be awarded accordingly.
 - The group leader should submit / upload a single set of files after collecting the individual contributions from group members.
- ☐ Evaluation will emphasize:
 - Code quality.
 - Result accuracy.
 - Result readability.
 - Result interpretation.
- ☐ Instances of plagiarism will result in mark deductions for all groups involved.

Academic Integrity

Honesty is primarily the responsibility of each student. The institute considers cheating to be a voluntary act for which there may be a reason, but for which there is no acceptable excuse. It is important to understand that collaborative learning is considered cheating unless specifically allowed for by the professor. The term cheating includes but is not limited to: plagiarism, receiving or knowingly supplying unauthorized information, using unauthorized material or sources, changing an answer after work has been graded and presenting it as improperly graded, illegally accessing confidential information through a computer, doing the assignment for another student or having another student do the assignment for you.